

MD Shahriar Hossain

Software Engineer · MERN / Node.js / Python
Open to internships & junior roles

shahriar1181@gmail.com
+880 1906127394
Mirpur-10, Dhaka, Bangladesh

LINKS

- github.com/Shahriar1638
- linkedin/md-shahriar-hossain-58b80a281
- shahriar1638.github.io/MyPortfolio

SKILLS

Languages

- JavaScript
- TypeScript
- Python
- C
- HTML5
- CSS3

Backend

- Node.js
- Express.js
- REST APIs
- JWT
- RBAC
- Stripe

Frontend

- React
- NextJs
- TanStack Query
- Tailwind CSS
- Shadcn/UI

Databases

- MongoDB
- MySQL
- PostgreSQL*
- Schema Design

DevOps / Tools

- Git
- GitHub
- Vercel
- Firebase
- Docker*

ML / DL

- PyTorch
- TensorFlow
- Scikit-learn
- NumPy
- Pandas

* familiar

EDUCATION

B.Sc. Computer Science

BRAC University
2022 - 2026 (Expected)
CGPA 3.29 / 4.00
120 / 124 credits completed
Thesis defense: Spring 2026

Higher Secondary Certificate

Monipur H.S. & College
2020 · GPA 4.17 / 5.00

SUMMARY

Software Engineer and CS graduate with hands-on experience in MERN, SQL, and secure API architectures. I'm Skilled at building secure data layers, integrating third party APIs and reliable backend systems while supporting responsive front ends that ensure seamless data flow and polished user interfaces. I have practical experience on Deep learning and Machine Learning projects through academy. Open to work in junior roles and internships where I can apply full-cycle development skills to real-world engineering challenges.

PROJECTS

SurveyHub — Full-Stack Survey Platform

[live](#) · [github](#)

React · Node.js · Express.js · MongoDB · JWT · Firebase Auth · TanStack Query · Shadcn/UI

- Developed a responsive full-stack survey management application featuring a live voting system and automated real-time data visualization using Recharts.
- Engineered a secure Role-Based Access Control (RBAC) architecture, managing complex state logic to provide dedicated dashboards and separate content moderation workflows.
- Strengthened security and performance by utilizing Axios interceptors for automated JWT injection, and leveraging TanStack Query to establish efficient caching and reduce redundant API calls.

BizNest — E-Commerce Platform

[live](#) · [github](#)

React · Node.js · Express.js · MongoDB · Mongoose · JWT · TanStack Query · Stripe

- Engineered a MERN-stack e-commerce marketplace with strict Role-Based Access Control (RBAC) to support separate Customer, Seller, and Admin workflows.
- Architected a decoupled React frontend using the Facade pattern and custom hooks, and applied an aggressive caching strategy via TanStack Query to drastically reduce redundant network requests.
- Implemented secure checkout flows via the Stripe API with server-side price recalculation, and hardened backend authentication against common vulnerabilities using Helmet.js, rate limiting, and Axios interceptors.

Grocery Hub — Database Management System

[github](#)

PHP · MySQL · Relational Database Design

- Designed a normalized relational schema (3NF) in MySQL covering products, orders, inventory, and user accounts, with foreign key constraints and indexed queries for data integrity.
- Built a full-cycle ordering workflow in PHP — browse, cart, checkout, inventory decrement — using parameterized queries throughout to prevent SQL injection.

RESEARCH

Multi-Omics Sarcoma Subtype Classification

2025 - Present

Undergraduate Thesis · [github.com/Shahriar1638/Multi-Omics-P3](#) · Gated Attention + DAE + MLP · PyTorch · Scikit-learn

- Designed a custom deep learning architecture combining Gated Attention, a Denoising Autoencoder (DAE), and MLP layers to classify 6 sarcoma subtypes from high-dimensional, class-imbalanced multi-omics data.
- Built end-to-end pipelines with Pandas and Scikit-learn for multi-omics feature alignment, normalization, and stratified splitting across a low-sample clinical dataset.